

Profitability & ROS Policy

Internal analytical guidance for interpreting Profit and Return on Sales (ROS) in the AI Business Analyst.

1. Profit definition

- Profit is provided as a field in the Orders data and should be treated as the source profitability measure.
- The available Orders fields do not provide a documented cost field or a complete profit-calculation formula. The AI Analyst should therefore not invent a cost or recompute Profit from Sales and Discount.
- Negative Profit is a valid analytical outcome and should be investigated rather than treated as a data error automatically.

2. Return on Sales

- Overall ROS is calculated as $\text{SUM(Profit)} \div \text{SUM(Sales)} \times 100$.
- For a row-level or order-level ROS, $\text{Profit} \div \text{Sales} \times 100$ can be used where Sales is non-zero.
- Average order-level ROS and overall ROS are different measures and should not be used interchangeably.
- For grouped analysis, calculate ROS using the aggregate numerator and denominator for the same group unless the question explicitly asks for the average of individual ROS values.

3. Profitability reasoning

- High gross sales do not necessarily imply high profit or high ROS.
- When investigating low profitability, examine discount, product mix, quantity, unit price and other available dimensions before forming a conclusion.
- A negative ROS means aggregate Profit is negative relative to gross Sales for the selected population.

4. AI Analyst guidance

- Prefer evidence such as contribution to total profit, ROS, order count and discount exposure when explaining profitability.
- Separate observed facts from possible explanations.
- If the data cannot establish a causal explanation, state that the factor is associated with the outcome rather than claiming causation.